Part 2: Planetary Orbits, Kepler's Laws, and How Our Planets Look from Afar

Sakarias Frette[1]

[1] *AST2000 Rerun, University of Oslo*

(Dated: November 7, 2026)

We compute the orbits of the 8 planets around our 1.187 $M_\odot$ star, both analytically and with a leapfrog integrator, over 21 orbits of the home planet. Using the exact solution of Kepler's equation as the reference, we choose the time step from a measured error curve. The worst position error is 0.18%, below the 1% the official check allows, and `ast2000tools` accepted the orbits. The simulation obeys Kepler's second law to 2.0×10^{-9} and the third law to 7.0×10^{-6}, and energy along the orbit satisfies the vis-viva law to 1.5×10^{-11}. Letting the star move as well, we model what an observer far away would measure. The heaviest planet, 6, moves the star at 58.8 m/s, and from a noisy radial-velocity curve we recover $m \sin i$ to -0.6%. Its transit, with a depth of 0.02190, gives the planet's radius and its density, 1922 kg/m^3 (true: 1934). A detectability survey shows that an observer could find the gas giants, but hardly any of the rocky planets. As an extra, we measure general-relativistic perihelion precession: our code reproduces Mercury's 42.98″ per century.

I. INTRODUCTION

Before we can fly anywhere we need to know where the planets are, at any time, accurately enough to aim a spacecraft. This part [1] builds that knowledge.

- Analytical and numerical orbits, verified by `ast2000tools` [2] (Section II).
- Kepler's laws as tests of the simulation (Section III).
- The same system seen from outside: the star's wobble, radial velocities and transits (Section IV).

We use astronomical units throughout: AU, yr and $M_\odot$, in which $G = 4\pi^2\,\mathrm{AU}^3\,\mathrm{yr}^{-2} M_\odot^{-1}$. Solar system seed: 42716.

II. ORBITS

A. Theory

For a planet that only feels a fixed star of mass M, the orbit is a conic section [3],

$$r(\theta) = \frac{a(1-e^2)}{1 + e\cos(\theta - \omega)}, \tag{1}$$

with semi-major axis a, eccentricity e and perihelion angle ω. The position at a given *time* follows from Kepler's equation $M_{\mathrm{anom}} = E - e\sin E$. Here the mean anomaly grows linearly in time, and E is the eccentric anomaly. We solve it with Newton's method for all planets and times at once. It is our exact reference.

B. Method

We integrate all planets together with the kick–drift–kick leapfrog scheme,

$$\begin{aligned} \mathbf{v}_{i+1/2} &= \mathbf{v}_i + \tfrac{1}{2}\mathbf{a}(\mathbf{r}_i)\Delta t, \\ \mathbf{r}_{i+1} &= \mathbf{r}_i + \mathbf{v}_{i+1/2}\Delta t, \\ \mathbf{v}_{i+1} &= \mathbf{v}_{i+1/2} + \tfrac{1}{2}\mathbf{a}(\mathbf{r}_{i+1})\Delta t. \end{aligned} \tag{2}$$

Leapfrog is symplectic [4], so its energy error stays bounded instead of drifting. For a central force it conserves angular momentum exactly. Its error is a phase lag that grows linearly in time. We measure it as the distance from the exact Kepler position, divided by the distance to the star, which is what the API checks.

C. Results

Figures 1 and 2 show the system. The home planet orbits at $a = 0.687$ AU with $e = 0.0013$ and a period of 0.523 yr. Periods range from 0.068 yr for the innermost planet to 19.7 yr for the outermost. The three outer planets are gas giants with densities below 2 g/cm^3. The rocky planets have densities of 4–6 g/cm^3.

The time-step study (Fig. 4) shows the expected Δt^2 convergence. The minimum allowed 10 000 steps per year would give a worst error of 2.8%, which fails the check, because the innermost planet completes an orbit in only 0.068 yr. We therefore use 40,000 steps per year. Over 10.98 yr this gives a worst error of 0.18% (planet 7, Fig. 5). Energy is conserved to 2.7×10^{-8} and angular momentum to 1.5×10^{-13} (Fig. 6).

`ast2000tools` accepted the orbits (passed). Its own Kepler solver crashes at certain sample times, so we handed it every 2. sample. The API then saved the exact orbits, which all later parts use.

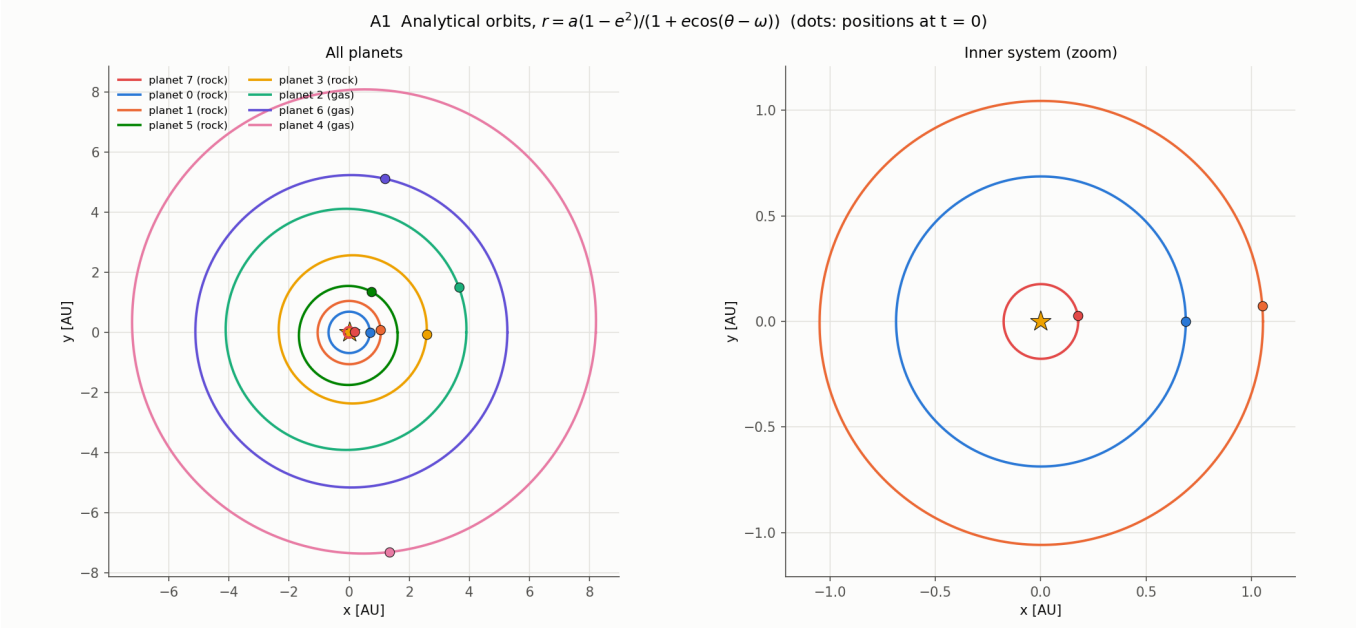


FIG. 1. Analytical orbits from Eq. (1), with the planets' positions at $t = 0$.

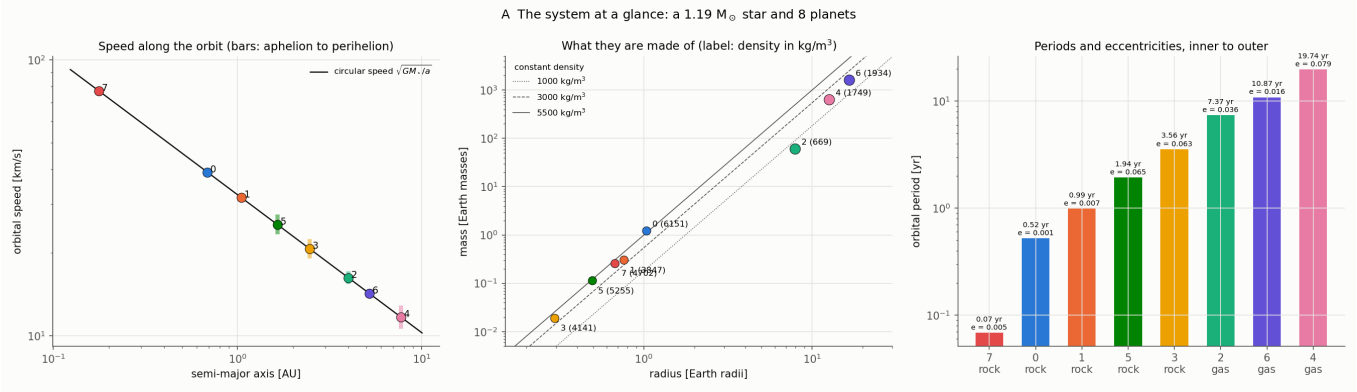


FIG. 2. The system at a glance. Orbital speed against distance; mass against radius with lines of constant density; periods and eccentricities.

III. KEPLER'S LAWS

A. Second law

The line from the star to a planet sweeps area at the rate $\mathrm{d}A/\mathrm{d}t = |\mathbf{r} \times \mathbf{v}|/2 = h/2$. This is constant because the angular momentum h is conserved for any central force. We measured the area swept in 0.05 of an orbit near perihelion and near aphelion (Fig. 7). For the most eccentric planet (planet 4, $e = 0.0795$) it moves 1.1719 times faster at perihelion. Yet the two areas agree to 2.0×10^{-9}.

B. Vis-viva

Energy conservation along an ellipse gives $v^2 = GM(2/r - 1/a)$. Figure 8 shows that the simulation follows it to 1.5×10^{-11}. This is an independent check of the integrator.

C. Third law

Kepler's law with a fixed star, $P^2 = a^3/M$, matches our measured periods to 7.0×10^{-6} (Fig. 9). Newton's version $P^2 = a^3/(M + m)$ differs by a factor of about $1 - m/2M$. That is only 0.206% even for the heaviest planet (6), and our fixed-star model cannot see it. The two-body simulation in Section IV does.

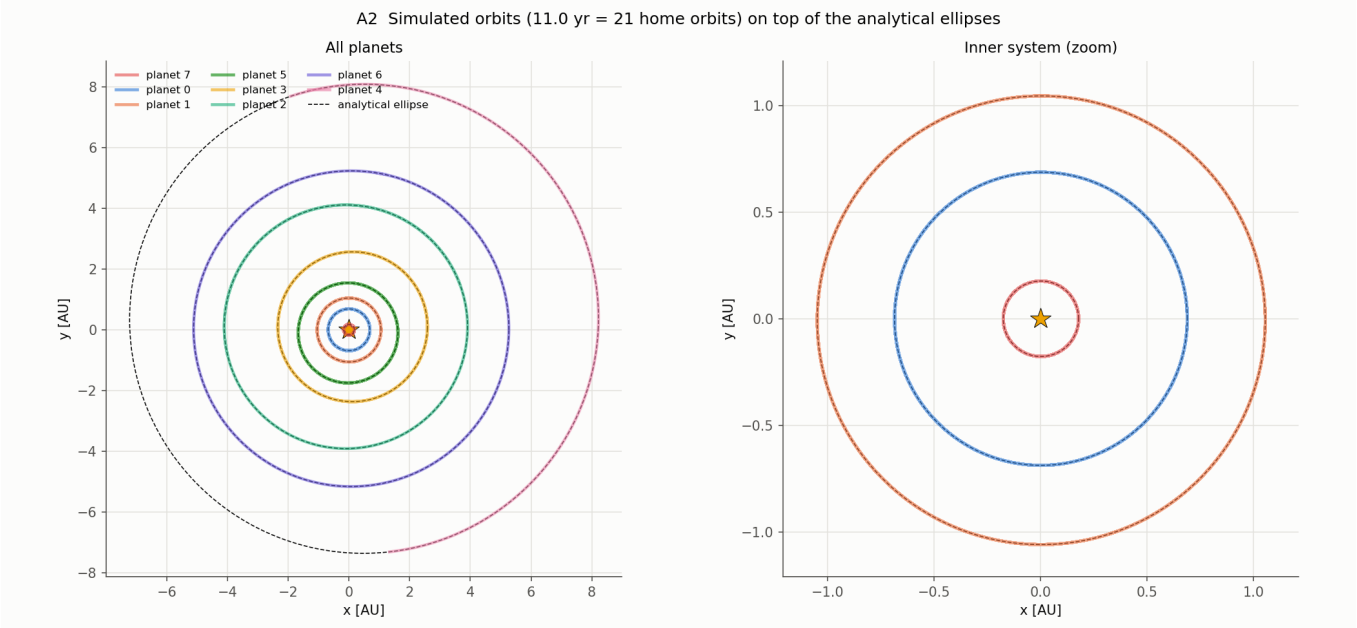


FIG. 3. Simulated orbits (color) on top of the analytical ellipses (dashed).

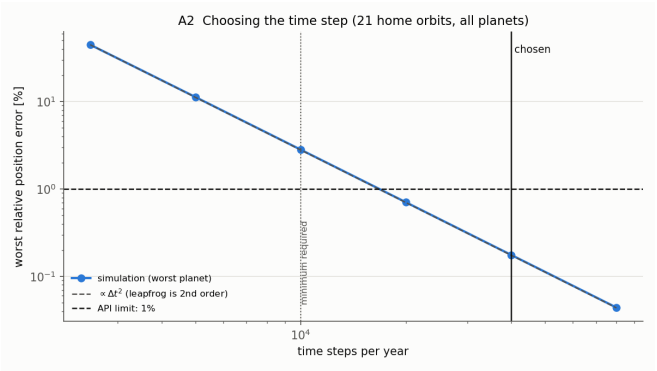


FIG. 4. Worst relative position error against the number of steps per year. Leapfrog converges as Δt^2.

IV. SEEN FROM AFAR

A. The star's wobble

The planet that moves the star the most has the largest $m/\sqrt{a}$. Here that is planet 6, with $m = 4.899 \times 10^{-3} M_\odot$ at $a = 5.19\,\mathrm{AU}$. We integrate star and planet together with leapfrog in the centre-of-mass frame. The star then moves on a small ellipse of radius 3,182,064 km at 58.8 m/s (Fig. 10). The two-body energy $\frac{1}{2}\hat{\mu}v^2 - GM\hat{\mu}/r$ is conserved to 3.5×10^{-10}. The measured period, 10.7793 yr, matches Newton's $P^2 = a^3/(M+m)$ (10.7793 yr). Kepler's a^3/M would give 10.8015 yr, which is 8.1 days too long.

B. Radial velocity

An observer in the orbital plane's direction at inclination $i = 70°$ measures $v_r = v_{\mathrm{pec}} + v_{\star,x} \sin i$. We add a peculiar velocity of 2500 m/s and Gaussian noise of 11.2 m/s, and take 1500 measurements. We then analyse the curve as an outsider would. The mean gives v_{pec}, and a smoothed curve gives the period and amplitude. For $m \ll M$ on a circular orbit, these give the minimum mass

$$m \sin i = \frac{M^{2/3}\, v_\star\, P^{1/3}}{(2\pi G)^{1/3}}. \tag{3}$$

By eye we read $P = 10.83\,\mathrm{yr}$ and $v_\star = 55.0\,\mathrm{m/s}$, so $m \sin i = 4.577 \times 10^{-3} M_\odot$ (Fig. 11). The true value is 4.604×10^{-3}, so we are off by -0.6%. A least-squares sine fit gives 4.602×10^{-3}.

C. Transit

Seen edge-on, the planet blocks a fraction $(R_p/R_\star)^2$ of the star's light. We model the flux as a trapezoid with linear ingress and egress, add 1×10^{-4} noise, and fit (Fig. 12). The depth 0.02190 gives $R_p = 106,356\,\mathrm{km}$. Combined with the relative speed from the radial-velocity analysis, the 4.19-hour ingress gives 108,033 km. The true radius is 106,337 km. With the mass from radial velocity, the density is $1922\,\mathrm{kg/m^3}$ (true: 1934): a gas giant.

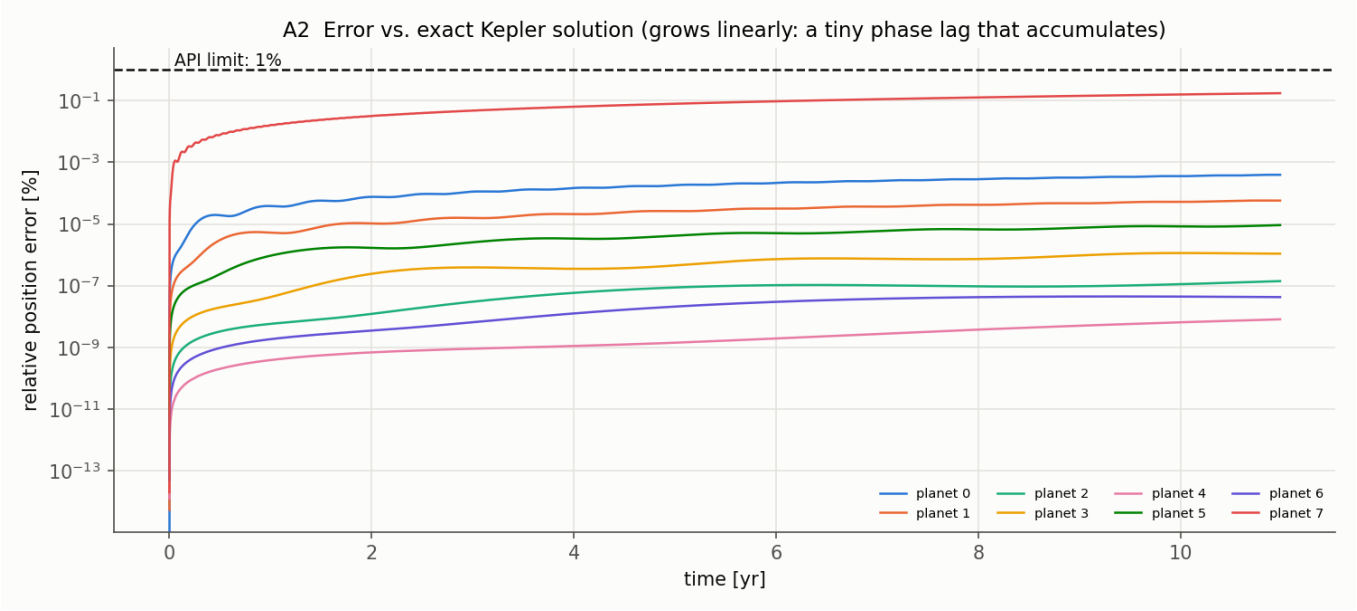


FIG. 5. Position error against the exact Kepler solution for every planet: a phase lag that grows linearly.

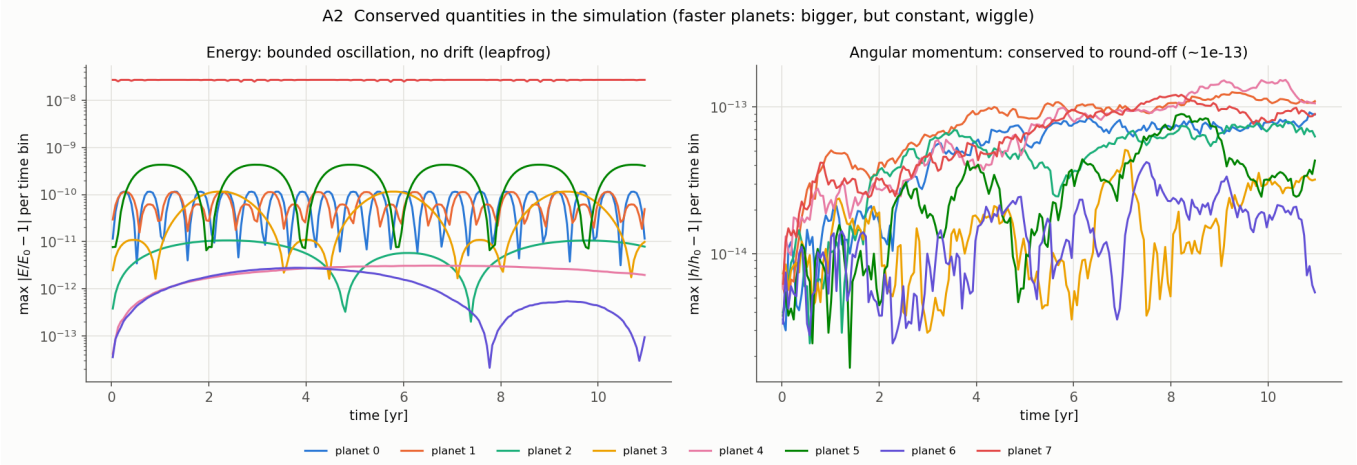


FIG. 6. Largest relative change in energy and angular momentum per time bin.

D. Several planets

With all three gas giants (planets 2, 6, 4) in a full N-body run of 45 yr, the star's velocity is the sum of three wobbles (Fig. 13). We search the noisy curve with a Lomb–Scargle periodogram [5]. At each round we subtract the strongest sine and search again ("prewhitening"), and at the end we fit all three sines jointly. The true periods are 7.37, 10.87, 19.74 yr, and the joint fit gives 5.74, 10.39, 20.23 yr. It recovers the two heaviest giants to within a few percent but misplaces the lightest one. That planet (2) moves the star by only 2.3 m/s, much less than the noise per point. Over a span of only a few orbits it also blends with the harmonics of the two big signals. Separating it would need longer or less noisy observations.

E. Could they find us?

Figure 14 puts every planet against the noise of our synthetic observations. In radial velocity, only planets 2, 4, 6 stand out at 5σ after averaging. The home planet moves the star by only 11.5 cm/s. In transit, planets 0, 2, 4, 6 are deep enough once a hundred points in transit are averaged. But an observer must also be lined up with the orbit, which for the home planet happens with probability $R_\star/a = 0.70\%$.

V. EXTRA: GENERAL RELATIVITY

In Schwarzschild spacetime the orbit equation gains a term $3GMu^2/c^2$ (with $u = 1/r$). This is a central force

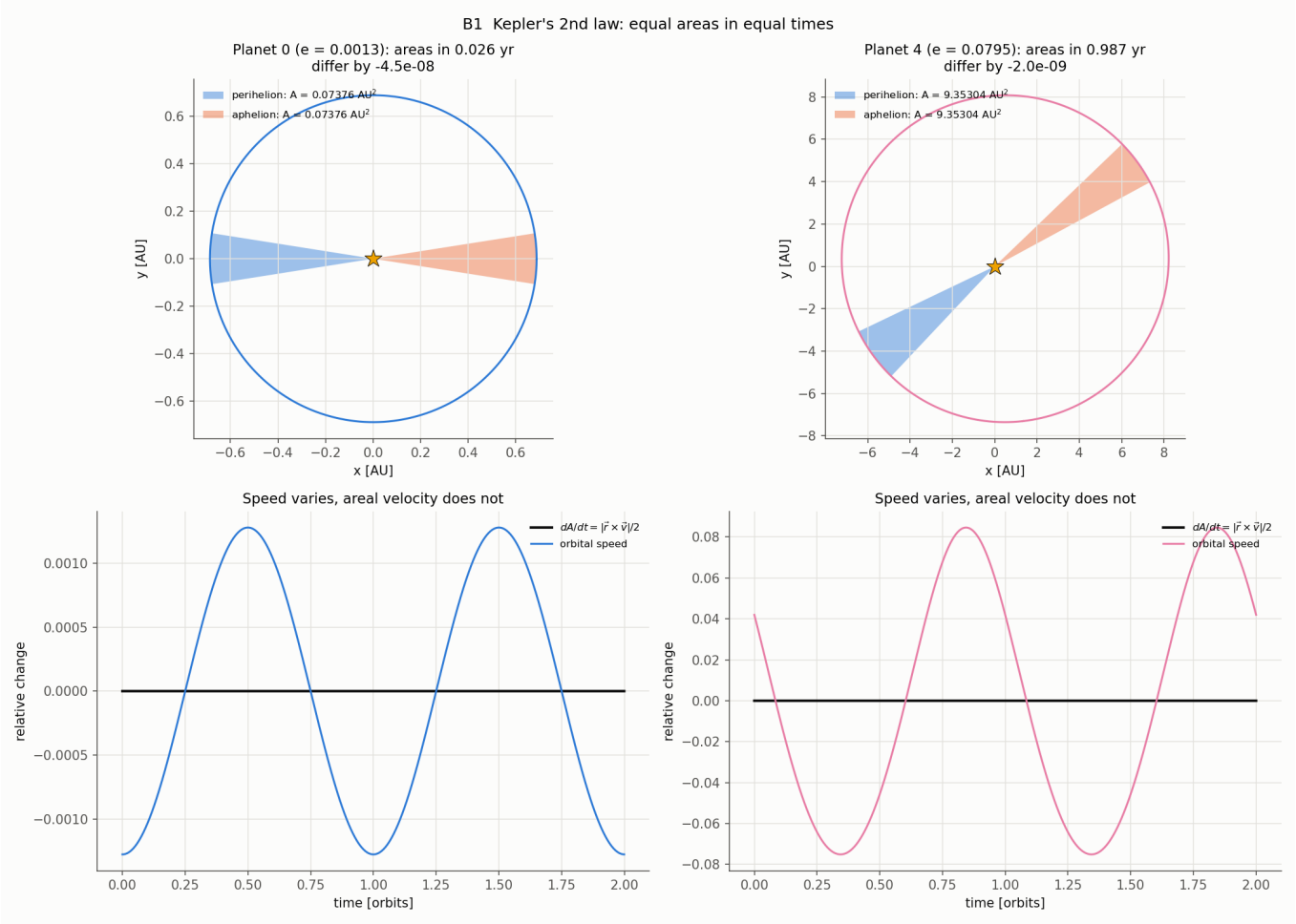


FIG. 7. Kepler's second law. Equal areas swept in equal times near perihelion and aphelion (top). The areal velocity stays constant while the speed varies (bottom).

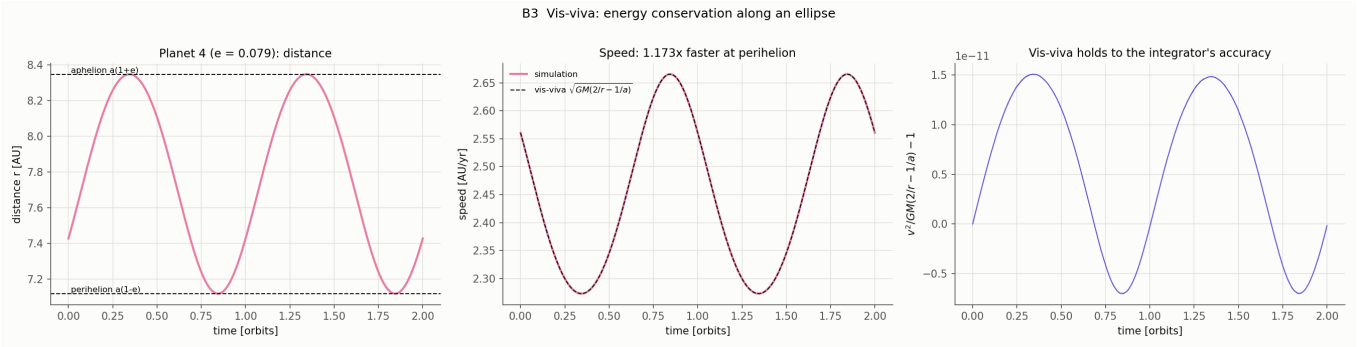


FIG. 8. Distance, speed and the vis-viva relation for the most eccentric planet.

$\mathbf{a} = -(GM/r^2)(1 + 3h^2/c^2r^2)\hat{\mathbf{r}}$, so leapfrog still conserves h exactly. GR rotates the orbit by about 10^{-7} rad per orbit, which is smaller than leapfrog's own numerical precession. We therefore run every orbit twice with identical steps, Newtonian and relativistic, and subtract, so the numerical error cancels. We track the perihelion with the Laplace–Runge–Lenz vector.

The method reproduces Mercury's 42.98″ per century (Fig. 15). For our innermost planet (7) it gives 376.9″ per century (predicted: 376.7). Over the 20 checked orbits, GR moves the home planet by 1385 km. That is far less than the planet-planet forces a full N-body run adds (Fig. 16).

That run also revealed something important. Planets

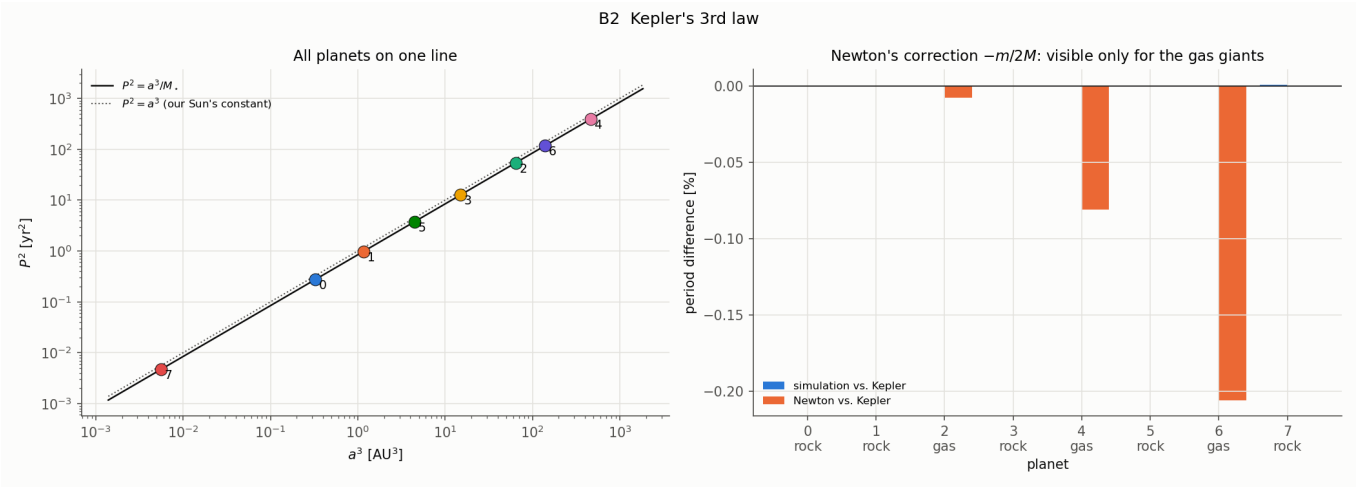


FIG. 9. Kepler's third law for all planets, and the size of Newton's correction.

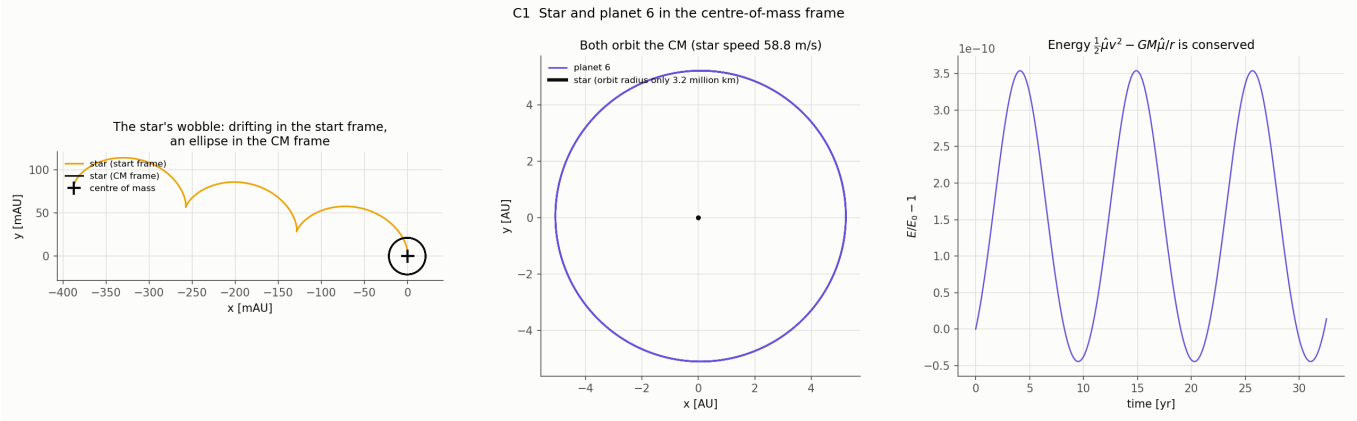


FIG. 10. The star's wobble around the centre of mass, the two orbits, and energy conservation.

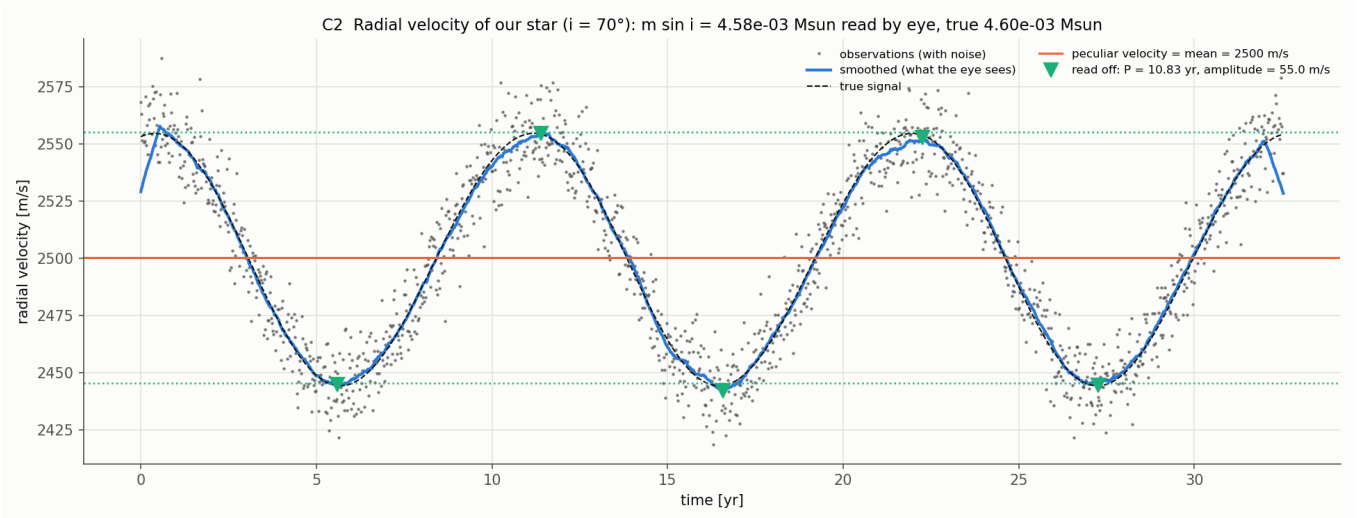


FIG. 11. Noisy radial-velocity curve, the smoothed curve an observer would read, and the extracted period and amplitude.

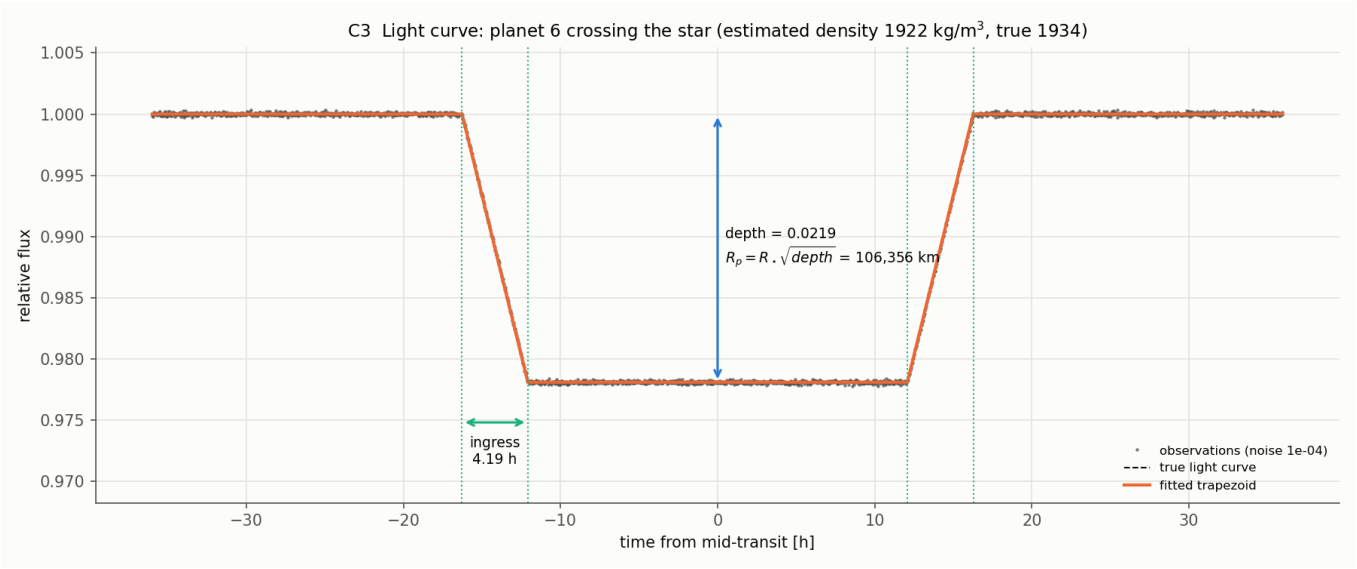


FIG. 12. Light curve of planet 6 crossing the star, with the fitted trapezoid.

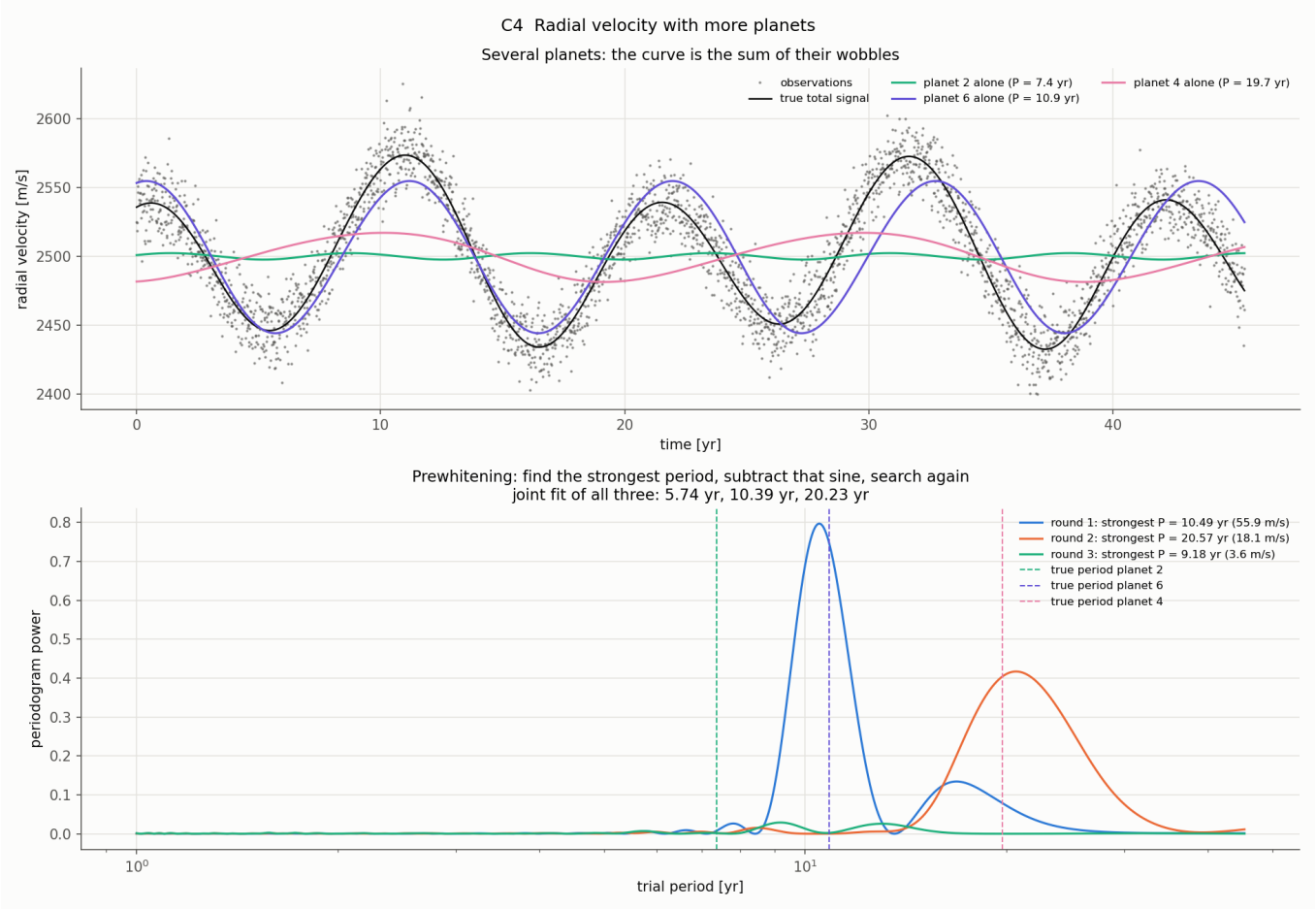


FIG. 13. Radial velocity with three giants (top) and the prewhitening periodograms (bottom).

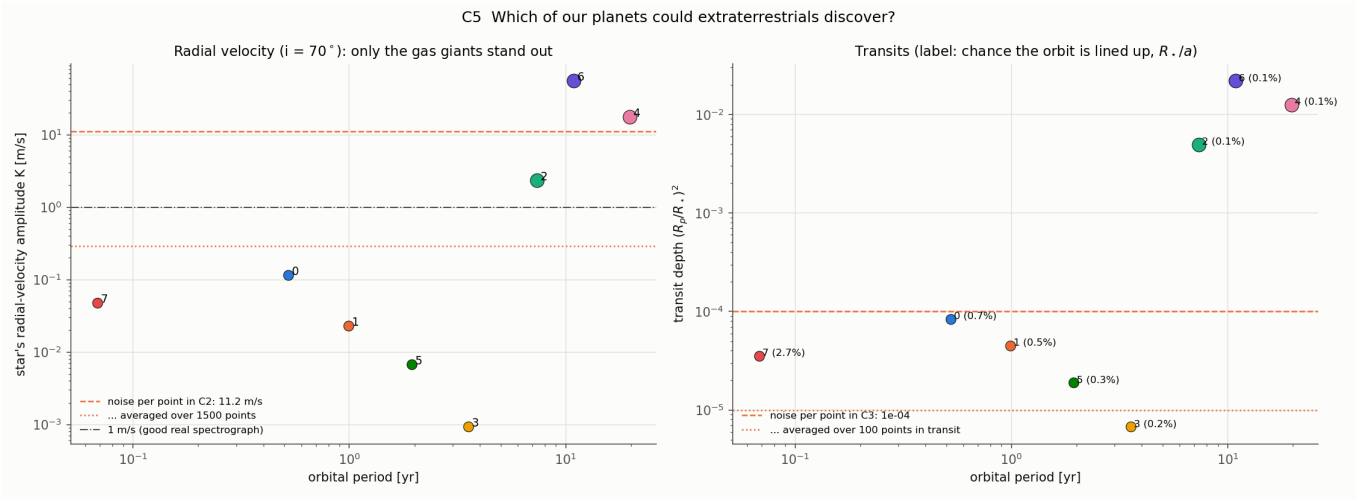


FIG. 14. Detectability of every planet by radial velocity (left) and transit (right), against the noise of our synthetic data.

2 and 6 pass only 0.019 AU apart at $t = 5.2$ yr, so the generated system is dynamically unstable. The API, whose truth is Kepler ellipses, ignores this.

VI. DISCUSSION AND CONCLUSION

Choosing the time step from a measured error, instead of a guess, was the key step. The innermost planet sets the requirement, and it needs four times more steps than the minimum allowed. Kepler's equation turned out to be valuable well beyond this part. It gives exact positions at any time, and Parts 3–5 use it instead of interpolating our own orbits.

The outside view shows how biased planet detection is. Heavy, close-in planets dominate radial velocity, large planets dominate transits, and Earth-like planets hide below the noise. Our system would look like a three-giant system to most distant observers.

ACKNOWLEDGMENTS

`pro_2.py` reproduces every figure and number; `rv_curve.txt` and `light_curve.txt` are the synthetic data for the partner group.

[1] AST2000 teaching team, AST2000 spacecraft project, parts 1–9, Institute of Theoretical Astrophysics, University of Oslo (2023).

[2] F. K. Hansen and L. F. Engedal, ast2000tools: Python package for the AST2000 project, https://frodekh.github.io/ast2000tools/html/index.html (2023).

[3] F. K. Hansen, AST2000 lecture notes, Institute of Theoretical Astrophysics, University of Oslo (2017).

[4] E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration*, 2nd ed. (Springer, Berlin, 2006).

[5] N. R. Lomb, Least-squares frequency analysis of unequally spaced data, Astrophysics and Space Science **39**, 447 (1976).

Appendix A: Mission log

Part 2 key numbers (seed 42716)	
Star mass	$1.187\,M_\odot$
Home orbit a, e, P	0.687 AU, 0.0013, 0.523 yr
Steps per year	40,000
Worst position error	0.18%
API verification	passed
Kepler 2nd / 3rd law	2.0×10^{-9} / 7.0×10^{-6}
RV $m \sin i$ error	-0.6%
Transit density	1922 kg/m^3

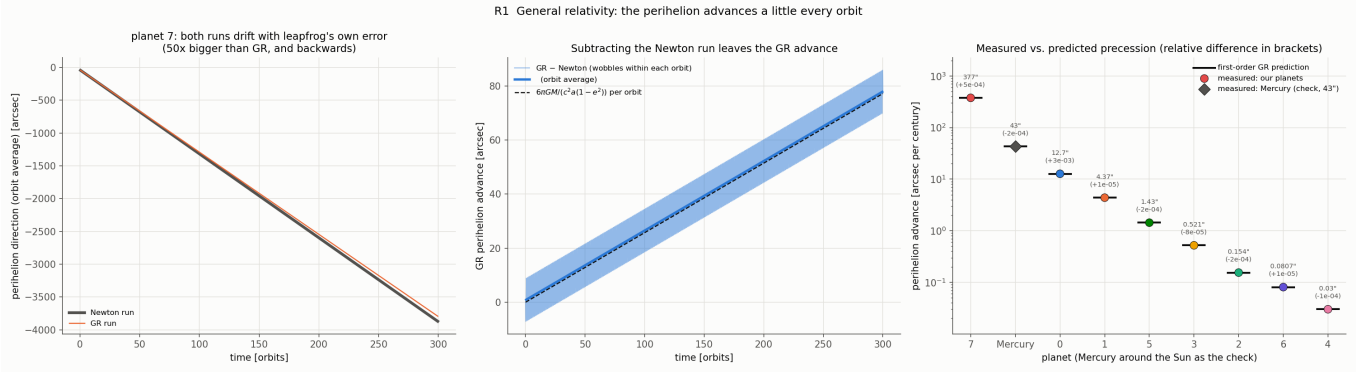


FIG. 15. Measured against predicted GR perihelion precession for all planets and Mercury.

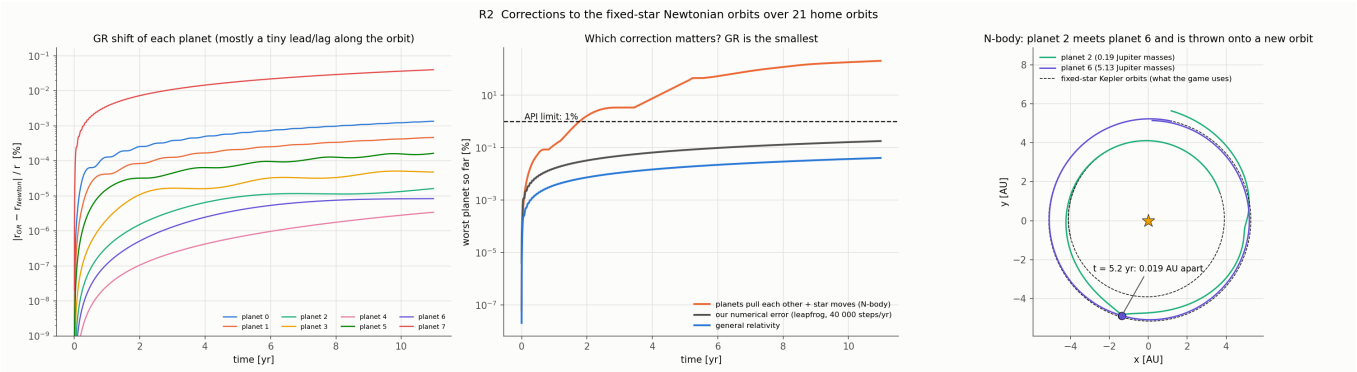


FIG. 16. Size of the GR and N-body corrections compared with our numerical error.

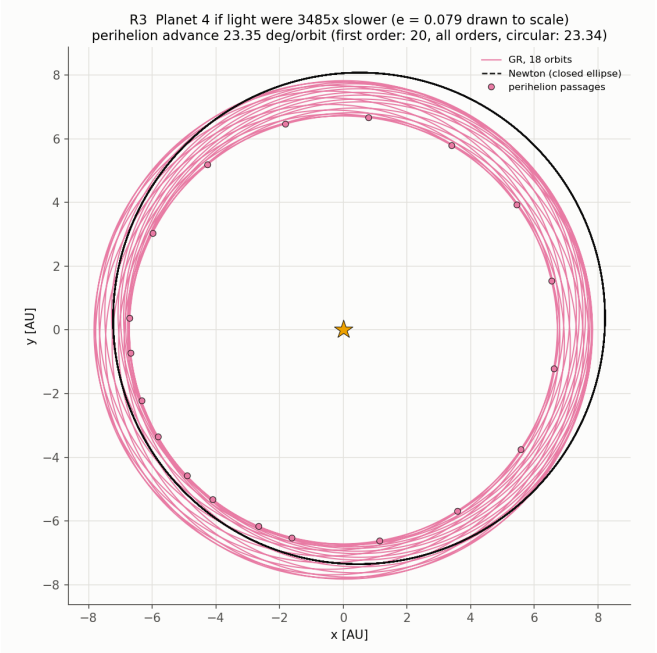


FIG. 17. GR precession made visible by lowering c: the orbit becomes a rosette.